

Simulations (draft)

We simulate raw categorical ratings with $R = 6$ raters and $C = 5$ categories, using $T_i \sim \text{Categorical}(p)$ with empirical p taken from `misskappa::dat.fleiss1971`. Complete-data truth is approximated by a large Monte Carlo run ($n = 30000$) under each DGP.

Moderate sample size: $n = 600$ with $B = 40$ replications. Large sample size: $n_{big} = 6000$ with $B_{big} = 15$ replications.

Main: Conger’s kappa (nominal/identity loss)

Table 1: Simulation bias and Monte Carlo SD for Conger’s κ_C (nominal/identity loss).

DGP	Method	κ_C (truth)	Bias (SD), $n = 600$	Bias (SD), $n = 6000$
A	AC	0.8449	-0.0025 (0.0109)	-0.0009 (0.0038)
A	IPW	0.8449	-0.0016 (0.0126)	-0.0013 (0.0041)
A	Gwet	0.8449	-0.0025 (0.0109)	-0.0009 (0.0038)
B	AC	0.6938	0.1118 (0.0174)	0.1116 (0.0062)
B	IPW	0.6938	0.0033 (0.0256)	0.0023 (0.0088)
B	Gwet	0.6938	0.1124 (0.0173)	0.1121 (0.0062)
C	AC	0.6681	0.1159 (0.0145)	0.1197 (0.0045)
C	IPW	0.6681	0.1160 (0.0145)	0.1197 (0.0046)
C	Gwet	0.6681	0.1159 (0.0145)	0.1197 (0.0045)

Note. Rating model: $T_i \sim \text{Categorical}(p)$; conditional on T_i , rater j reports T_i with probability ρ_{ij} and otherwise guesses from p . Missingness indicators $M_{ij} \in \{0, 1\}$ are independent across (i, j) given the specified mechanism. DGP A: exchangeable raters ($\rho_{ij} = 0.92$) and MCAR with rater-specific $\pi_j = \Pr(M_{ij} = 1)$ given by $(0.95, 0.90, 0.80, 0.65, 0.50, 0.35)$. DGP B: non-exchangeable raters with $\rho_{ij} = \rho_j$ where $\rho_j = (0.98, 0.96, 0.92, 0.85, 0.72, 0.60)$ and MCAR with $\pi_j = (0.95, 0.85, 0.70, 0.45, 0.25, 0.15)$ (higher-skill raters observed more often). DGP C: category-dependent difficulty with $\rho_{ij} = 0.92 m(T_i)$ where $m = (1.05, 1.00, 0.95, 0.85, 0.70)$, and MAR with $\pi_c = \Pr(M_{ij} = 1 \mid T_i = c) = (0.95, 0.90, 0.80, 0.55, 0.25)$. Main table uses nominal (identity) loss; Appendix table uses linear loss $\ell(a, b) = |a - b|/(C - 1)$ (treating categories as ordered).

Appendix (draft)

Conger’s kappa (linear/absolute loss)

Fleiss’ kappa (identity loss)

DGP	Method	truth	Bias (SD), $n = 600$	Bias (SD), $n = 6000$
A	AC	0.8449	-0.0025 (0.0109)	-0.0009 (0.0038)
A	IPW	0.8449	-0.0017 (0.0126)	-0.0013 (0.0041)
A	Gwet	0.8449	-0.0025 (0.0109)	-0.0009 (0.0038)
B	AC	0.6937	0.1117 (0.0174)	0.1116 (0.0062)
B	IPW	0.6937	0.0031 (0.0256)	0.0023 (0.0088)
B	Gwet	0.6937	0.1123 (0.0173)	0.1121 (0.0062)
C	AC	0.6681	0.1159 (0.0146)	0.1197 (0.0046)
C	IPW	0.6681	0.1159 (0.0145)	0.1197 (0.0046)
C	Gwet	0.6681	0.1159 (0.0146)	0.1197 (0.0046)

Table 2: Simulation bias and Monte Carlo SD for Conger’s κ_C (linear/absolute loss).

DGP	Method	κ_C (truth)	Bias (SD), $n = 600$	Bias (SD), $n = 6000$
A	AC	0.8452	-0.0054 (0.0156)	-0.0054 (0.0041)
A	IPW	0.8452	-0.0046 (0.0171)	-0.0055 (0.0049)
A	Gwet	0.8452	-0.0054 (0.0157)	-0.0054 (0.0041)
B	AC	0.6924	0.1078 (0.0151)	0.1105 (0.0051)
B	IPW	0.6924	-0.0012 (0.0308)	0.0007 (0.0073)
B	Gwet	0.6924	0.1093 (0.0151)	0.1119 (0.0051)
C	AC	0.6635	0.1289 (0.0159)	0.1268 (0.0049)
C	IPW	0.6635	0.1290 (0.0159)	0.1268 (0.0049)
C	Gwet	0.6635	0.1289 (0.0159)	0.1268 (0.0049)

Brennan–Prediger (identity loss)

DGP	Method	truth	Bias (SD), $n = 600$	Bias (SD), $n = 6000$
A	AC	0.8484	-0.0024 (0.0107)	-0.0010 (0.0038)
A	IPW	0.8484	-0.0016 (0.0122)	-0.0014 (0.0041)
A	Gwet	0.8484	-0.0024 (0.0107)	-0.0010 (0.0038)
B	AC	0.6993	0.1107 (0.0169)	0.1102 (0.0061)
B	IPW	0.6993	0.0037 (0.0251)	0.0023 (0.0087)
B	Gwet	0.6993	0.1107 (0.0169)	0.1102 (0.0061)
C	AC	0.6716	0.1168 (0.0144)	0.1204 (0.0045)
C	IPW	0.6716	0.1169 (0.0143)	0.1204 (0.0045)
C	Gwet	0.6716	0.1168 (0.0144)	0.1204 (0.0045)